

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 6

1 Complexity Theory (tg508)

Let **Factor** be the *decision* problem where given a pair of integers (x, k) , the goal is to decide whether x has a factor smaller than k . Let **Factoring** be the *search* problem, where given an integer x , the goal is to output a prime factorisation of x . (In the following, carefully note the distinction between **Factor** and **Factoring**.)

coNP

- (a) Prove that **Factor** $\in$ NP $\cap$ coNP. [5 marks]

Answer: To see that **Factor** $\in$ NP, simply note that a prime $p \leq k$ that divides n is a succinct certificate. To see that **Factor** $\in$ coNP, recall that **Primes** $\in$ P, and hence the prime factorisation of x , which does not include any primes less than k , is a certificate that (x, k) is not in **Factor**.

NP Completeness
/ reductions

- (b) Prove that if $P = \text{NP} \cap \text{coNP}$, then there exists a polynomial-time algorithm for **Factoring**. [7 marks]

Answer: Assume $P = \text{NP}$. Hence **Factor** $\in$ P and we can decide in polynomial time whether (x, k) has a factor smaller than k . Note that x has $O(\log|x|)$ factors. Now perform a binary search with respect to k , using the polynomial-time algorithm for **Factor** to find a factor. Now, divide by the factor and repeat until all prime factors are found.

Quantum
Complexity

- (c) Define the class BQP. Is **Factoring** $\in$ BQP? [4 marks]

Answer: The class BQP is the class of all decision problems that can be solved, with probability at least $2/3$, via a uniform family of quantum circuits. While Shor's algorithm can factorise a prime number, by definition, BQP consists of *decision* problems only, and hence does not contain the *search* problem **Factoring**.

Quantum
Complexity

- (d) Show that a quantum (BQP) algorithm for a problem P , which is correct with probability $2/3$ over the measurement, can be amplified into a quantum algorithm for P , which is correct with probability $1 - o(1)$ over the measurement. [4 marks]

Answer: Error reduction can be performed by repeating the algorithm t times and ruling according to the majority of the answer, then carefully applying the Chernoff bound and performing the necessary calculations to show that the distribution is exponentially concentrated around the mean.
